

First Booster-Free Integrated Tunable Laser achieving 100-mW Output Power over 111-nm Tuning Range

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Abstract We demonstrate a novel hybrid III-V/SiN laser, first to deliver 100 mW fiber-coupled output power without a booster-SOA over a record tuning range (111 nm), side-mode-suppression-ratio greater than 60 dB and a linewidth lower than 3 kHz, validated with a 64-QAM sub-system.

1. Introduction

III-V semiconductor materials provide efficient optical gain, while silicon-based photonic platforms offer low-loss waveguides and flexible wavelength-selective circuits. Their complementary properties have enabled hybrid integrated lasers combining a III-V gain section with a silicon or silicon nitride external cavity [1-2]. Designing active chips alongside tuning elements – such as sampled gratings [3], Mach-Zehnder interferometers [4], and micro-ring Vernier filters [5] – make these architectures attractive for ultra-wide wavelength tuning applications. Moreover, chip-to-chip coupling loss in hybrid external-cavity lasers directly increases the cavity loss and limits the achievable output power. Various integration approaches, including free-space optics [6], spot-size converters [7], precision butt-coupling [8], and flip-chip assembly [9] have been developed to improve coupling efficiency. However, these methods generally require accurate mode matching and stringent lateral, vertical, and angular alignment. These strict constraints can be significantly alleviated through photonic wire bonding (PWB) [10]. By forming a three-dimensional polymer waveguide between independently fabricated chips, PWB enables broadband mode transformation and relaxed alignment tolerances along with low coupling loss.

However, maintaining high output power across a broad tuning range remains challenging, because gain and passive-circuit filtering characteristics vary with wavelength and power. Indeed, the state-of-the-art of devices providing ultra-wide spectral coverage [5] [11-14] generally operate at relatively low output powers, whereas high-power demonstrations (Fig. 1) are typically restricted to

narrower wavelength ranges [6, 15]. Output powers of approximately 20 dBm have previously been achieved across the C-band only by employing an additional booster SOA [16], which increases the footprint, power consumption, and assembly complexity.

In this work, we demonstrate a photonic-wire-bonded III-V/SiN tunable laser achieving a 111-nm tuning range while maintaining more than 100 mW in-fiber output power without an additional booster amplifier thanks to a new III-V RSOA design and low-loss cavity. To the best of our knowledge, this is the first demonstration of a booster-free integrated tunable laser delivering 20-dBm power in fiber over such a wide tuning range (Fig. 1). Furthermore, the benefit of this new TLS has been confirmed by comparing a 64-QAM sub-system performance with a commercial μ TILA.

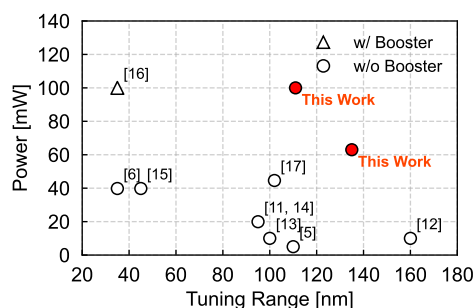


Fig. 1. Output power vs tuning range of tunable lasers.

2. Hybrid laser design

The schematic of the hybrid laser is shown in Fig. 2. A reflective semiconductor optical amplifier (RSOA) is coupled to an external silicon-nitride-based cavity. The RSOA comprises a 1.25-mm-long ridge waveguide incorporating InGaAlAs

multiple quantum wells (MQWs) as the gain medium. To enable high output power, the active section is designed with a low optical confinement factor of approximately 1% and a wide ridge (3.25 μm). The resulting large optical mode area reduces the optical intensity in the gain region and increases the saturation output power. Since the low confinement factor also reduces the modal gain, InGaAlAs MQWs are selected for their large material gain to thereby maintain a high net gain in the RSOA across C+L+U band. The ridge waveguide was angled at 6° relative to the RSOA facet to reduce intra-cavity reflection.

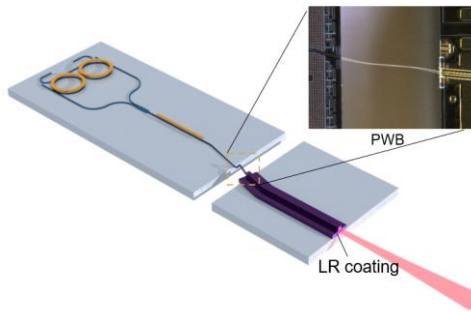


Fig. 2. Schematic of the ECL laser coupled by PWB; Inset: Microscopy of the PWB.

The external cavity is implemented on a silicon nitride photonic integrated circuit (PIC) and incorporates a Vernier based tunable reflector formed by two micro-ring resonators (MRRs) connected in a serial add-drop configuration (Manhattan configuration). A broadband, low-loss 1×2 multimode interferometer (MMI) is used to split and recombine the optical field between the bus waveguide and the dual-ring reflector. The two MRRs are designed with close free-spectral-ranges (FSR) such as the resulting Vernier FSR exceeds 135 nm and thereby enabling continuous wavelength coverage across the C+L-band. A relatively low power-coupling coefficient (around 5%) is selected to achieve a favorable compromise between filter selectivity and optical loss. 1-dB total reflection-path loss is hence measured for the SiN external-cavity circuit.

At the coupling facet, the SiN waveguide is tapered and optimized with a facet angle of 15.7 degree, matching that of the RSOA in order to facilitate chip-to-chip coupling and suppress parasitic back-reflections. Optical coupling between the RSOA and the SiN PIC is realized using a carefully optimized PWB as illustrated in Fig. 2.

Anti-reflection (AR) coating is applied to the RSOA facet which is connected to the PWB to reduce parasitic reflections at the chip-to-chip interface.

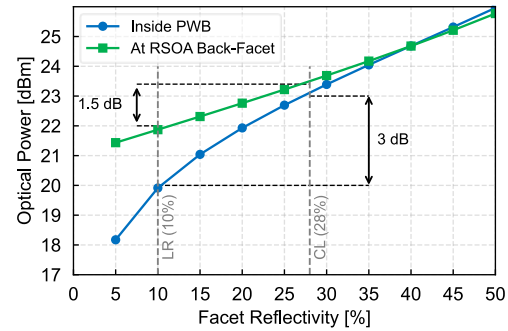


Fig. 3. Calculation of the power inside PWB and at RSOA back-facet with 20 dBm output power (ignoring laser nonlinearities).

The RSOA back facet, which serves as the laser output, is coated with a low-reflectivity (LR) coating of approximately 10%, providing efficient power extraction while keeping the optical power transmitted through the PWB low enough.

As shown in Fig. 3, the 10%-reflectivity design allows the power within the PWB to be lowered by 3 dB in comparison to a cleaved facet. In addition, the lower facet reflectivity reduces the photon density by 1.5 dB within the RSOA, thereby reducing the laser nonlinearities for 20 dBm output power.

3. Experimental results and discussion

The demonstrated laser was packaged as a module with thermoelectric cooler (TEC), free-space collimator, isolator, and coupled to the fiber. The total coupling loss from laser output to the fiber is estimated to be 1dB. The performance was characterized at 35°C controlled temperature. While tuning the MRRs and phase section with integrated micro-heaters, the RSOA drive current was varied (~600 to ~775 mA) to ensure a constant 20-dBm output power across the spectral range. Power measurements were recorded directly at the module's output fiber using a calibrated optical spectrum analyzer (OSA). The corresponding optical spectra are shown in Fig. 4(a). A wavelength tuning range of 111 nm (13.0 THz), from 1545 to 1656 nm, was achieved while maintaining an output power of 20 dBm and a side-mode suppression ratio (SMSR) exceeding 60 dB. The achieved spectral range is narrower than the 135-nm tunable reflector FSR because of the thermal impairments induced by larger driving current required at shorter wavelengths. Indeed, with the target output power reduced to 18 dBm, the tuning range was further extended to 135 nm (Fig. 4(b)). Measured light-current (L-I) characteristics provided the threshold current and slope efficiency. Comparing these measured thresholds to values computed from the RSOA

gain and SiN filter loss yields an estimated PWB coupling loss of 1 to 2 dB across the tuning range. Low-loss external cavity, combined with RSOA's high saturation power (>17 dBm), enables the high booster-free output power.

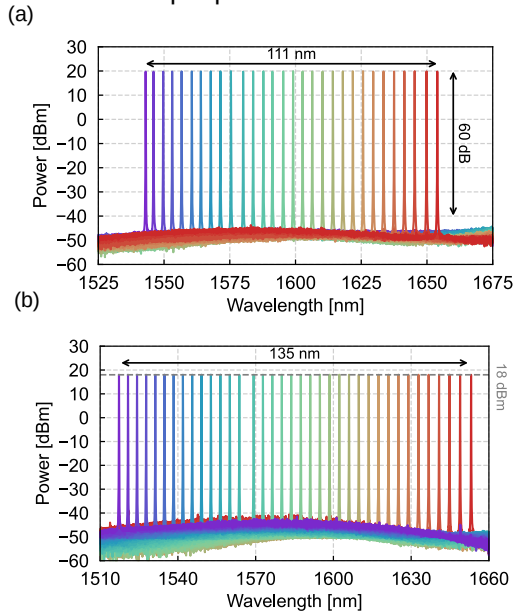


Fig. 4. The measured (a) 20 dBm and (b) 18 dBm output power (in-fiber) spectra with 111 nm and 135 nm tuning range respectively at 35°C.

Furthermore, the intrinsic/Lorentzian linewidth of the laser was evaluated using a digital coherent receiver method [18] based on optical heterodyne beating with a commercial laser (EXFO T100S-HP). Digital signal processing of the captured complex amplitude yields to the frequency modulation (FM) noise spectrum. The intrinsic linewidth is then extracted from the floor of the FM noise spectrum; this floor is typically observed within the 5 to 50 MHz range. Each datapoint in Fig. 5 is the average value over 20 acquisitions of the FM noise spectrum. Benefiting from the low PWB coupling loss and the high finesse of the SiN filter, the laser achieves an intrinsic linewidth ranging from 0.7 to 3 kHz at an output power of 20 dBm, and from 1 to 4 kHz at 18 dBm (Fig. 5(a)). Note that the measurement spectral span was constrained solely by the wavelength coverage of the reference laser. The intrinsic linewidth at 1576 nm was further characterized across various output powers by varying the RSOA current (Fig. 5(b)). By comparing these measurements with our simulations, the linewidth enhancement factor of the RSOA is expected to remain between 1 and 3 across the evaluated current range.

Furthermore, the proposed TLS is compared to a high-performance commercial micro integrated

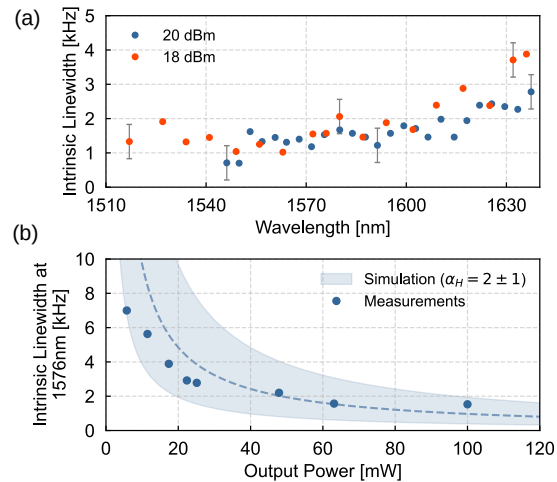


Fig. 5. (a): Measured linewidth at 20 dBm and 18 dBm in-fiber output power and 35°C. (b): Measured linewidth at 1576 nm compared to simulation vs. laser output power.

tunable laser assembly (μ TLA) [19] using external cavity and modulated at 64GBaud with PCS-64QAM transmission over 40 km SSMF fiber. Digital subcarrier multiplexing (DSCM) is increased from 1 to 64 to highlight laser linewidth impact. At receiver, VOA is placed before the EDFA to allow equal OSNR for proposed TLS vs μ TLA, and standard DSP chain was used. The benefit of 20 dBm power vs 14.5 dBm μ TLA is shown in Fig. 6. In particular, 3-dB SNR is improved in single carrier. In DSCM 64, with VOA turned on for equal OSNR, 1 dB gain is obtained owing to our narrower TLS linewidth.

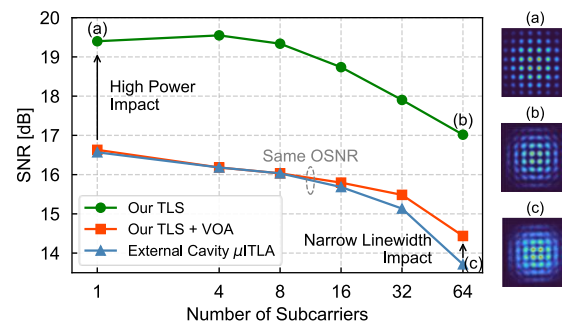


Fig. 6. Measured SNR and constellations vs DSCM at 64GBaud post 40km transmission.

4. Conclusion

Gain medium, facet coating, tunable reflector, and integration scheme have been carefully co-designed. The first, booster-free integrated tunable laser with simultaneously 20-dBm output power beyond 100-nm spectral range and kHz-level intrinsic linewidth, was achieved. The benefit of our high-power and low-linewidth TLS to 64QAM system performance has been confirmed.

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